

BULGAN BURENKHAIKHAAN, Mongolia

WMO: 442650

Lat: 46.0947N

Lon: 91.5522E

Elev: 1186

StdP: 87.87

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.0	-31.2	-36.4	0.1	-33.7	-33.8	0.2	-30.9	6.2	-13.6	4.9	-11.2	0.1	50	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	31.8	17.5	30.2	16.9	28.7	16.4	19.9	28.7	18.7	27.6	17.6	26.2	2.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.7	24.9	15.3	12.6	23.7	14.1	11.6	23.2	62.9	28.7	58.3	27.6	54.3	26.2	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.2	-37.2	34.9	4.3	1.6	-40.3	36.1	-42.8	37.1	-45.3	38.0	-48.4	39.2		
(5)			-37.0	20.8	4.5	2.4	-40.3	22.5	-42.9	23.9	-45.4	25.2	-48.7	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-20.1	-14.8	-3.5	7.4	14.1	19.9	21.6	19.5	13.0	4.5	-7.0	-16.7	(6)
(7)		DBStd	15.35	5.56	6.10	6.97	5.20	4.04	3.12	2.64	3.02	4.08	4.64	6.74	5.39	(7)
(8)		HDD10.0	3694	933	694	419	104	12	0	0	0	18	177	510	827	(8)
(9)		HDD18.3	5749	1191	928	678	327	140	19	4	22	165	429	760	1086	(9)
(10)		CDD10.0	1228	0	0	0	27	138	297	360	295	107	5	0	0	(10)
(11)		CDD18.3	242	0	0	0	0	8	66	106	58	4	0	0	0	(11)
(12)		CDH23.3	2802	0	0	0	4	139	713	1119	736	89	0	0	0	(12)
(13)		CDH26.7	869	0	0	0	0	22	207	399	229	13	0	0	0	(13)
(14)	Wind	WSAvg	1.6	0.5	0.8	1.5	2.5	2.6	2.5	2.1	1.9	1.8	1.5	1.0	0.6	(14)
(15)	Precipitation	PrecAvg	83	4	3	4	5	6	12	15	13	4	5	8	5	(15)
(16)		PrecMax	140	27	14	14	21	36	46	37	70	16	19	39	19	(16)
(17)		PrecMin	44	0	0	0	0	0	1	3	0	0	0	0	0	(17)
(18)		PrecStd	25	5	3	5	5	8	11	9	14	5	5	8	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.6	3.0	16.1	23.9	29.2	32.7	35.0	33.0	28.5	20.9	11.5	-0.8	(19)
(20)			MCWB	-6.5	-1.9	6.8	12.3	15.3	18.3	20.1	18.1	15.9	11.5	4.7	-4.5	(20)
(21)		2%	DB	-6.5	-0.1	13.1	21.1	26.8	30.8	32.4	31.0	25.9	18.0	8.2	-4.0	(21)
(22)			MCWB	-8.1	-3.0	5.3	10.6	14.2	16.6	18.4	17.2	14.5	9.2	2.7	-6.4	(22)
(23)		5%	DB	-8.9	-2.6	10.3	18.9	24.8	29.1	30.6	29.3	23.7	15.8	5.5	-6.2	(23)
(24)			MCWB	-10.2	-4.8	3.9	9.4	13.4	16.2	17.4	16.7	13.3	8.1	1.0	-7.9	(24)
(25)		10%	DB	-11.3	-5.1	7.6	16.9	22.7	27.5	28.9	27.5	21.8	13.4	2.7	-8.4	(25)
(26)			MCWB	-12.3	-6.7	2.4	8.3	12.2	15.6	16.7	15.8	12.3	6.7	-0.8	-9.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.9	-0.9	7.7	13.4	17.6	20.7	22.1	20.6	18.6	12.9	5.3	-4.1	(27)
(28)			MCDWB	-4.1	1.5	14.3	22.4	27.3	30.7	32.1	29.6	26.3	19.8	10.5	-1.4	(28)
(29)		2%	WB	-8.1	-2.8	5.8	11.5	15.6	18.6	20.4	19.2	15.8	10.1	3.2	-6.2	(29)
(30)			MCDWB	-6.7	-0.3	12.3	19.6	24.7	28.1	29.8	28.0	23.7	16.7	8.2	-4.3	(30)
(31)		5%	WB	-10.2	-4.8	4.3	10.1	14.1	17.5	19.2	17.9	14.2	8.5	1.4	-7.8	(31)
(32)			MCDWB	-8.9	-2.6	10.1	17.7	23.0	26.4	27.9	27.1	21.9	14.7	5.5	-6.1	(32)
(33)		10%	WB	-12.3	-6.8	2.4	8.8	12.8	16.5	17.8	16.7	12.7	6.9	-0.4	-9.6	(33)
(34)			MCDWB	-11.3	-5.1	7.2	16.1	21.5	24.8	26.1	25.4	20.7	12.9	2.9	-8.3	(34)
(35)	Mean Daily Temperature Range	MDBR	13.4	14.5	13.7	13.1	12.5	12.0	12.3	13.6	14.7	14.2	12.5	12.1	(35)	
(36)		5% DB	MCDBR	13.9	15.2	15.8	16.6	15.7	15.0	15.5	16.4	17.2	17.8	16.0	13.4	(36)
(37)			MCWBR	12.3	12.3	9.8	9.3	8.1	6.8	6.8	7.4	8.9	11.1	10.8	11.0	(37)
(38)			MCDBR	13.8	14.7	15.4	15.4	14.3	12.7	14.0	14.8	15.9	17.0	15.0	13.0	(38)
(39)		5% WB	MCWBR	12.2	12.1	9.7	9.2	8.2	7.1	7.5	7.9	9.4	11.2	10.6	10.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.263	0.298	0.339	0.385	0.379	0.390	0.381	0.359	0.330	0.317	0.286	0.267	(40)	
(41)		taud	2.271	2.187	2.154	2.117	2.209	2.254	2.315	2.356	2.387	2.340	2.317	2.251	(41)	
(42)		Ebn at Noon	839	868	880	868	886	875	879	887	885	840	800	785	(42)	
(43)		Edh at Noon	79	106	128	146	138	133	124	114	101	90	75	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.97	3.05	4.47	5.80	6.85	7.05	6.89	6.15	5.09	3.33	2.04	1.58	(44)	
(45)		RadStd	0.10	0.12	0.28	0.32	0.40	0.43	0.29	0.20	0.24	0.19	0.17	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page